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global_data = {

    # --- Attribute Names (unchanged) ---
    "attribute_1": "total_cost",
    "attribute_2": "material_continuity",
    "attribute_3": "copper_oz_weight",
    "attribute_4": "wind_exposure_level",
    "attribute_5": "building_type_tier",

    # --- Attribute Directions (unchanged) ---
    "direction_attribute_1": "negative",
    "direction_attribute_2": "positive",
    "direction_attribute_3": "positive",
    "direction_attribute_4": "positive",
    "direction_attribute_5": "positive",

    # -----
    # REVISED ATTRIBUTE BOUNDS
    # Designed to hold for any future wind drip product, any region, any era.
    # Current 8-product dataset range noted in brackets for reference.
    # -----

    # ATTRIBUTE 1 — total_cost (USD per 10-ft section)
    # Current dataset: [$3.82 – $50.32]
    # Lower floor: Thinnest possible Al (0.016") on narrowest 4" stock, bulk-order
    # commodity price in the lowest-cost manufacturing market imaginable:
    # ~0.48 kg Al × $2.50 = $1.20 material + ~$0.30 forming/overhead = ~$1.50.
    # Set to $1.00 — slightly below even that floor to safely catch any future
    # hyper-low-cost product.
    # Upper ceiling: Thickest credible copper (32 oz) on a wide 12" developed
    # stock, bespoke heritage fabrication in the most expensive labor market:
    # 10 ft × 1.0 ft × 32 oz/ft² = 20 lbs = 9.07 kg × $9.50 = $86.17 material
    # + $60 bespoke fabrication/overhead = ~$146. Set to $200 — provides
    # generous headroom for any future copper price spike or ultra-premium spec.
    "attribute_1_min": 1.00,
    "attribute_1_max": 200.00,

    # ATTRIBUTE 2 — material_continuity (binary: 0 or 1)
    # This is a hard binary flag by definition. 0 = prior aluminum; 1 = prior copper.
    # There is no physically meaningful value outside {0, 1}. Bounds are exact
    # and correct for all possible future datasets — no exaggeration needed or possible.
    "attribute_2_min": 0,
    "attribute_2_max": 1,

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# ATTRIBUTE 3 — copper_oz_weight (oz/ft²)
# Current dataset: [0 – 20]
# Lower floor: 0 is exact and immovable — a pure aluminum product carries
# zero copper by definition. Cannot go below 0.
# Upper ceiling: ASTM B370 standard production grades run:
# 10, 12, 14, 16, 20, 24, 32 oz/ft².
# 32 oz/ft² is the heaviest standard architectural copper sheet produced.
# Any wind drip heavier than 32 oz would be a non-standard bespoke casting,
# not a rolled sheet product — outside the product category entirely.
# Set to 32 to span the full ASTM B370 production range.
"attribute_3_min": 0,
"attribute_3_max": 32,

# ATTRIBUTE 4 — wind_exposure_level (ordinal integer scale 1–5)
# This is a defined ordinal scale anchored to physical wind severity categories:
# 1 = sheltered/inland, 2 = moderate suburban, 3 = high coastal/open,
# 4 = very high hurricane/typhoon zone, 5 = extreme direct major-storm coast.
# The scale is fully bounded by its own definition. No product or region can
# logically fall outside 1–5. Bounds are exact and universal.
"attribute_4_min": 1,
"attribute_4_max": 5,

# ATTRIBUTE 5 — building_type_tier (ordinal integer scale 1–4)
# This is a defined ordinal scale anchored to specification tiers:
# 1 = standard residential, 2 = premium residential,
# 3 = commercial/institutional/luxury, 4 = heritage/landmark/estate.
# As with attribute 4, the scale is self-bounding by definition. No wind drip
# product exists outside these four tiers. Bounds are exact and universal.
"attribute_5_min": 1,
"attribute_5_max": 4,

# --- Material Prices (unchanged) ---
"copper_price_per_ton": 9500,
"aluminum_price_per_ton": 2500,
"copper_price_per_kg": 9.5,
"aluminum_price_per_kg": 2.5
}

# =====
# WIND DRIP (F-TYPE DRIP EDGE FLASHING) — FULL SYSTEM INPUT DATA
# Product basis: F-Type (Gutter Apron) Wind Drip Edge, 10 ft section
# Developed width: 7.2" (confirmed K&M spec: 4" face × 10 ft = 7.2" stock)
# =====

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# TIER 2 — REGIONAL DATA
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# 4 entries. Regions MUST match product_data exactly.
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# Weights previously established in the regional attribute-weight analysis.
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# -----
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regional_data = [
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{
  "region": "United States",
  "copper_product_market_share": 0.15,    # ~15% of wind drip market — pr
  "aluminum_product_market_share": 0.85,   # ~85% — dominates

  "weight_attribute_1": 0.25,              # Total cost — significant but not singular; c
  "weight_attribute_2": 0.20,              # Material continuity — galvanic m
  "weight_attribute_3": 0.20,              # Copper oz weight — IRC/ASCE
  "weight_attribute_4": 0.20,              # Wind exposure — hurricane zone,
  "weight_attribute_5": 0.15,              # Building type — estate homes + ins
},
{
  "region": "China",
  "copper_product_market_share": 0.07,     # ~7% — niche: prestig
  "aluminum_product_market_share": 0.93,    # ~93% — volume-driven new

  "weight_attribute_1": 0.35,              # Total cost — dominant filter; copper m
  "weight_attribute_2": 0.15,              # Material continuity — thin legacy Cu
  "weight_attribute_3": 0.20,              # Copper oz weight — formalizing via J
  "weight_attribute_4": 0.20,              # Wind exposure — typhoon coastal z
  "weight_attribute_5": 0.10,              # Building type — prestige tier is real but t
},
{
  "region": "India",
  "copper_product_market_share": 0.04,     # ~4% — very niche: temples, heritage
  "aluminum_product_market_share": 0.96,    # ~96% — PMAY mass housing

  "weight_attribute_1": 0.40,              # Total cost — highest weight across all re
  "weight_attribute_2": 0.10,              # Material continuity — minimal mode
  "weight_attribute_3": 0.15,              # Copper oz weight — IS 875 Pt.3 win
  "weight_attribute_4": 0.25,              # Wind exposure — Bay of Bengal +
  "weight_attribute_5": 0.10,              # Building type — temples, heritage
},
{
  "region": "Europe",
  "copper_product_market_share": 0.30,     # ~30% — strongest Cu penetration
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    "aluminum_product_market_share": 0.70,    # ~70% — standard resident

    "weight_attribute_1": 0.15,                # Total cost — lowest weight; whole-life cost
    "weight_attribute_2": 0.25,                # Material continuity — heritage preservation
    "weight_attribute_3": 0.15,                # Copper oz weight — EN standards apply;
    "weight_attribute_4": 0.15,                # Wind exposure — Atlantic/Alpine exposure real
    "weight_attribute_5": 0.30,                # Building type — HIGHEST of all regions; li
  },
]

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# TIER 3 — PRODUCT DATA
# 8 entries: 1 Cu-dominant + 1 Al-dominant per region.
# Regions MUST match regional_data exactly.
#
# Mass derivation (all using 7.2" / 0.6 ft developed width, 10 ft length):
# Area per section = 10 ft × 0.6 ft = 6.00 ft²
#
# Cu 20 oz: 6.00 ft² × 20 oz/ft² = 120 oz = 7.50 lbs = 3.402 kg
# Cu 16 oz: 6.00 ft² × 16 oz/ft² = 96 oz = 6.00 lbs = 2.722 kg
# Al 0.024": Vol = 10×0.6×(0.024/12) ft³ = 0.012 ft³ = 339.8 cm³; 339.8×2.70 = 917g = 0.918 kg
# Al 0.019": Vol = 10×0.6×(0.019/12) ft³ = 0.0095 ft³ = 269.0 cm³; 269.0×2.70 = 727g = 0.726
kg
#
# Computed total costs (material + non-material):
# US Cu: 3.402×9.50 + 15.00 = 32.32 + 15.00 = $47.32 [validates vs. K&M retail $53.95 →
14% margin]
# US Al: 0.918×2.50 + 7.00 = 2.30 + 7.00 = $9.30
# EU Cu: 3.402×9.50 + 18.00 = 32.32 + 18.00 = $50.32
# EU Al: 0.918×2.50 + 9.00 = 2.30 + 9.00 = $11.30
# CN Cu: 2.722×9.50 + 6.00 = 25.86 + 6.00 = $31.86
# CN Al: 0.726×2.50 + 2.50 = 1.82 + 2.50 = $4.32
# IN Cu: 2.722×9.50 + 5.00 = 25.86 + 5.00 = $30.86
# IN Al: 0.726×2.50 + 2.00 = 1.82 + 2.00 = $3.82
# -----

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product_data = [

# — UNITED STATES —————
{
  "region":      "United States",
  "dominant material": "cu",
  "product":     "Copper F-Type Wind Drip Edge, 20 oz, 10 ft",

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# Mass
"copper_kg":    3.402,      # 20 oz/ft² × 6.00 ft² = 7.50 lbs = 3.402 kg (ASTM B370)
"aluminum_kg":  0,
"total_mass_kg": 3.402,
# Cost
"non_material_cost_per_unit": 15.00, # USD COGS excl. material; total $47.3
# Attributes
"attribute_2_value": 1,      # Previously copper installation (material continuity)
"attribute_3_value": 20,     # 20 oz/ft² — canonical heavy wind drip spe
"attribute_4_value": 4,      # Very high — Atlantic/Gulf hurricane zone, ASC
"attribute_5_value": 3,      # Commercial/institutional/estate — premium c
},
{
  "region":      "United States",
  "dominant material": "al",
  "product":      "Aluminum F-Type Wind Drip Edge, 0.024 in, 10 ft",
  # Mass
  "copper_kg":    0,
  "aluminum_kg":  0.918,     # 0.024" thick × 6.00 ft² area = 339.8 cm³ ×
  "total_mass_kg": 0.918,
  # Cost
  "non_material_cost_per_unit": 7.00, # USD COGS excl. material; total $9.30
  # Attributes
  "attribute_2_value": 0,     # Previously aluminum installation
  "attribute_3_value": 0,     # No copper content — 0 oz copper designation
  "attribute_4_value": 3,     # High — suburban/general wind-exposed resid
  "attribute_5_value": 1,     # Standard residential (mass-market application)
},
# — EUROPE —————
{
  "region":      "Europe",
  "dominant material": "cu",
  "product":      "Copper F-Type Wind Drip Edge, 20 oz, 10 ft",
  # Mass
  "copper_kg":    3.402,     # Same 20 oz canonical spec; European her
  "aluminum_kg":  0,
  "total_mass_kg": 3.402,
  # Cost
  "non_material_cost_per_unit": 18.00, # USD equiv. — higher EU labor + VAT-
  # Attributes
  "attribute_2_value": 1,     # Previously copper — strong continuity; h
  "attribute_3_value": 20,    # 20 oz/ft² — full spec for listed buildings and
  "attribute_4_value": 3,     # High — Atlantic/North Sea/Alpine coastal exp

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    "attribute_5_value": 4,          # Heritage/landmark/estate — listed buildings, p
  },
  {
    "region":      "Europe",
    "dominant material": "al",
    "product":      "Aluminum F-Type Wind Drip Edge, 0.024 in, 10 ft",
    # Mass
    "copper_kg":    0,
    "aluminum_kg":  0.918,          # Same 0.024" spec as US; European standar
    "total_mass_kg": 0.918,
    # Cost
    "non_material_cost_per_unit": 9.00,  # USD equiv. — higher EU labor base vs
    # Attributes
    "attribute_2_value": 0,          # Previously aluminum
    "attribute_3_value": 0,          # No copper content
    "attribute_4_value": 2,          # Moderate — typical interior European climate
    "attribute_5_value": 1,          # Standard residential (non-listed, general ho
  },

```

— CHINA —————

```

{
  "region":      "China",
  "dominant material": "cu",
  "product":      "Copper F-Type Wind Drip Edge, 16 oz, 10 ft",
  # Mass
  "copper_kg":    2.722,          # 16 oz/ft² × 6.00 ft² = 6.00 lbs = 2.722
  "aluminum_kg":  0,
  "total_mass_kg": 2.722,
  # Cost
  "non_material_cost_per_unit": 6.00,  # USD — lower Chinese manufac
  # Attributes
  "attribute_2_value": 1,          # Previously copper — prestige gove
  "attribute_3_value": 16,         # 16 oz/ft² — cost-moderated copper for
  "attribute_4_value": 3,          # High — coastal typhoon-adjacent zone (Gu
  "attribute_5_value": 3,          # Commercial/institutional — prestige civi
},

```

```

{
  "region":      "China",
  "dominant material": "al",
  "product":      "Aluminum F-Type Wind Drip Edge, 0.019 in, 10 ft",
  # Mass
  "copper_kg":    0,
  "aluminum_kg":  0.726,          # 0.019" thick × 6.00 ft² = 269.0 cm³ ×
  "total_mass_kg": 0.726,

```

```

# Cost
"non_material_cost_per_unit": 2.50,    # USD — low-cost mass manufactur
# Attributes
"attribute_2_value": 0,                # Previously aluminum
"attribute_3_value": 0,                # No copper content
"attribute_4_value": 2,                # Moderate — urban/suburban standard constru
"attribute_5_value": 1,                # Standard residential/commercial (mass
},

```

— INDIA —————

```

{
  "region":      "India",
  "dominant material": "cu",
  "product":     "Copper F-Type Wind Drip Edge, 16 oz, 10 ft",
  # Mass
  "copper_kg":   2.722,                # 16 oz/ft² — same cost-moderated Cu s
  "aluminum_kg": 0,
  "total_mass_kg": 2.722,
  # Cost
  "non_material_cost_per_unit": 5.00,  # USD — lowest labor cost base of the
  # Attributes
  "attribute_2_value": 1,                # Previously copper — temples, heritage
  "attribute_3_value": 16,               # 16 oz/ft² — standard copper wind drip for
  "attribute_4_value": 4,                # Very high — Bay of Bengal/Arabian Sea cy
  "attribute_5_value": 3,                # Commercial/institutional — heritage hotels,
},
{
  "region":      "India",
  "dominant material": "al",
  "product":     "Aluminum F-Type Wind Drip Edge, 0.019 in, 10 ft",
  # Mass
  "copper_kg":   0,
  "aluminum_kg": 0.726,                # Same 0.019" spec as China; cost-driven thin gauge
  "total_mass_kg": 0.726,
  # Cost
  "non_material_cost_per_unit": 2.00,  # USD — lowest non-material cost in da
  # Attributes
  "attribute_2_value": 0,                # Previously aluminum
  "attribute_3_value": 0,                # No copper content
  "attribute_4_value": 3,                # High — coastal/semi-coastal general resident
  "attribute_5_value": 1,                # Standard residential (PMAY affordable ho
},

```

]